Exercise 3.26 You have been enlisted to design a soda machine dispenser for your department lounge. Sodas are partially subsidized by the student chapter of the IEEE, so they cost only 25 cents. The machine accepts nickels, dimes, and quarters. When enough coins have been inserted, it dispenses the soda and returns any necessary change. Design an FSM controller for the soda machine. The FSM inputs are *Nickel*, *Dime*, and *Quarter*, indicating which coin was inserted. Assume that exactly one coin is inserted on each cycle. The outputs are *Dispense*, *ReturnNickel*, *ReturnDime*, and *ReturnTwoDimes*. When the FSM reaches 25 cents, it asserts *Dispense* and the necessary *Return* outputs required to deliver the appropriate change. Then it should be ready to start accepting coins for another soda.